

Introduction

I

Since the Greeks invented logic and geometry, the idea that all reasoning might be reduced to some kind of calculation—so that all arguments could be settled once and for all—has fascinated most of the Western tradition's rigorous thinkers. Socrates was the first to give voice to this vision. The story of artificial intelligence might well begin around 450 B.C. when (according to Plato) Socrates demands of Euthyphro, a fellow Athenian who, in the name of piety, is about to turn in his own father for murder: "I want to know what is characteristic of piety which makes all actions pious . . . that I may have it to turn to, and to use as a standard whereby to judge your actions and those of other men."¹ § Socrates is asking Euthyphro for what modern computer theorists would call an "effective procedure," "a set of rules which tells us, from moment to moment, precisely how to behave."²

Plato generalized this demand for moral certainty into an epistemological demand. According to Plato, all knowledge must be stateable in explicit definitions which anyone could apply. If one could not state his know-how in terms of such explicit instructions—if his knowing *how*

§Notes begin on p. 307. [Citations are indicated by a superior figure. Substantive notes are indicated by a superior figure and an asterisk.]

could not be converted into knowing *that*—it was not knowledge but mere belief. According to Plato, cooks, for example, who proceed by taste and intuition, and poets who work from inspiration, have no knowledge: what they do does not involve understanding and cannot be understood. More generally, what cannot be stated explicitly in precise instructions—all areas of human thought which require skill, intuition, or a sense of tradition—are relegated to some kind of arbitrary fumbling.

But Plato was not yet fully a cyberneticist (although according to Norbert Wiener he was the first to use the term), for Plato was looking for *semantic* rather than *syntactic* criteria. His rules presupposed that the person understood the meanings of the constitutive terms. In the *Republic* Plato says that Understanding (the rulelike level of his divided line representing all knowledge) depends on Reason, which involves a dialectical analysis and ultimately an intuition of the meaning of the fundamental concepts used in understanding. Thus Plato admits his instructions cannot be completely formalized. Similarly, a modern computer expert, Marvin Minsky, notes, after tentatively presenting a Platonic notion of effective procedure: “This attempt at definition is subject to the criticism that the *interpretation* of the rules is left to depend on some person or agent.”³

Aristotle, who differed with Plato in this as in most questions concerning the application of theory to practice, noted with satisfaction that intuition was necessary to apply the Platonic rules:

Yet it is not easy to find a formula by which we may determine how far and up to what point a man may go wrong before he incurs blame. But this difficulty of definition is inherent in every object of perception; such questions of degree are bound up with the circumstances of the individual case, where our only criterion *is* the perception.⁴

For the Platonic project to reach fulfillment one breakthrough is required: all appeal to intuition and judgment must be eliminated. As Galileo discovered that one could find a pure formalism for describing physical motion by ignoring secondary qualities and teleological considerations, so, one might suppose, a Galileo of human behavior might

succeed in reducing all semantic considerations (appeal to meanings) to the techniques of syntactic (formal) manipulation.

The belief that such a total formalization of knowledge must be possible soon came to dominate Western thought. It already expressed a basic moral and intellectual demand, and the success of physical science seemed to imply to sixteenth-century philosophers, as it still seems to suggest to thinkers such as Minsky, that the demand could be satisfied. Hobbes was the first to make explicit the syntactic conception of thought as calculation: "When a man *reasons*, he does nothing else but conceive a sum total from addition of parcels," he wrote, "for REASON . . . is nothing but reckoning. . . ."⁵

It only remained to work out the univocal parcels or "bits" with which this purely syntactic calculator could operate; Leibniz, the inventor of the binary system, dedicated himself to working out the necessary unambiguous formal language.

Leibniz thought he had found a universal and exact system of notation, an algebra, a symbolic language, a "universal characteristic" by means of which "we can assign to every object its determined characteristic number."⁶ In this way all concepts could be analyzed into a small number of original and undefined ideas; all knowledge could be expressed and brought together in one deductive system. On the basis of these numbers and the rules for their combination all problems could be solved and all controversies ended: "if someone would doubt my results," Leibniz said, "I would say to him: 'Let us calculate, Sir,' and thus by taking pen and ink, we should settle the question."⁷

Like a modern computer theorist announcing a program about to be written, Leibniz claims:

Since, however, the wonderful interrelatedness of all things makes it extremely difficult to formulate explicitly the characteristic numbers of individual things, I have invented an elegant artifice by virtue of which certain relations may be represented and fixed numerically and which may thus then be further determined in numerical calculation.⁸

Nor was Leibniz reticent about the importance of his almost completed program.

Once the characteristic numbers are established for most concepts, mankind will then possess a new instrument which will enhance the capabilities of the mind to far greater extent than optical instruments strengthen the eyes, and will supersede the microscope and telescope to the same extent that reason is superior to eyesight.⁹

With this powerful new tool, the skills which Plato could not formalize, and so treated as confused thrashing around, could be recuperated as theory. In one of his “grant proposals”—his explanations of how he could reduce all thought to the manipulation of numbers if he had money enough and time—Leibniz remarks:

the most important observations and turns of skill in all sorts of trades and professions are as yet unwritten. This fact is proved by experience when passing from theory to practice we desire to accomplish something. *Of course, we can also write up this practice, since it is at bottom just another theory more complex and particular. . . .*¹⁰

Leibniz had only promises, but in the work of George Boole, a mathematician and logician working in the early nineteenth century, his program came one step nearer to reality. Like Hobbes, Boole supposed that reasoning was calculating, and he set out to “investigate the fundamental laws of those operations of the mind by which reasoning is performed, to give expression to them in the symbolic language of a Calculus. . . .”¹¹

Boolean algebra is a binary algebra for representing elementary logical functions. If “*a*” and “*b*” represent variables, “.” represents “and,” “+” represents “or,” and “1” and “0” represent “true” and “false” respectively, then the rules governing logical manipulation can be written in algebraic form as follows:

$$\begin{array}{llll} a + a = a & a + 0 = a & a + 1 = 1 \\ a \cdot a = a & a \cdot 0 = 0 & a \cdot 1 = a \end{array}$$

Western man was now ready to begin the calculation.

Almost immediately, in the designs of Charles Babbage (1835), practice began to catch up to theory. Babbage designed what he called an “Analytic Engine” which, though never built, was to function exactly like a modern digital computer, using punched cards, combining logical

and arithmetic operations, and making logical decisions along the way based upon the results of its previous computations.

An important feature of Babbage's machine was that it was digital. There are two fundamental types of computing machines: analogue and digital. Analogue computers do not compute in the strict sense of the word. They operate by measuring the magnitude of physical quantities. Using physical quantities, such as voltage, duration, angle of rotation of a disk, and so forth, proportional to the quantity to be manipulated, they combine these quantities in a physical way and *measure* the result. A slide rule is a typical analogue computer. A digital computer—as the word *digit*, Latin for “finger,” implies—represents all quantities by discrete states, for example, relays which are open or closed, a dial which can assume any one of ten positions, and so on, and then literally *counts* in order to get its result.

Thus, whereas analogue computers operate with continuous quantities, all digital computers are discrete state machines. As A. M. Turing, famous for defining the essence of a digital computer, puts it:

[Discrete state machines] move by sudden jumps or clicks from one quite definite state to another. These states are sufficiently different for the possibility of confusion between them to be ignored. Strictly speaking there are no such machines. Everything really moves continuously. But there are many kinds of machines which can profitably be *thought of* as being discrete state machines. For instance in considering the switches for a lighting system it is a convenient fiction that each switch must be definitely on or definitely off. There must be intermediate positions, but for most purposes we can forget about them.¹²

Babbage's ideas were too advanced for the technology of his time, for there was no quick efficient way to represent and manipulate the digits. He had to use awkward mechanical means, such as the position of cogwheels, to represent the discrete states. Electric switches, however, provided the necessary technological breakthrough. When, in 1944, H. H. Aiken actually built the first practical digital computer, it was electromechanical, using about 3000 telephone relays. These were still slow, however, and it was only with the next generation of computers using vacuum tubes that the modern electronic computer was ready.

Ready for anything. For, since a digital computer operates with ab-

stract symbols which can stand for anything, and logical operations which can relate anything to anything, any digital computer (unlike an analogue computer) is a universal machine. First, as Turing puts it, it can simulate any other digital computer.

This special property of digital computers, that they can mimic any discrete state machine, is described by saying that they are *universal* machines. The existence of machines with this property has the important consequence that, considerations of speed apart, it is unnecessary to design various new machines to do various computing processes. They can all be done with one digital computer, suitably programmed for each case. It will be seen that as a consequence of this all digital computers are in a sense equivalent.¹³

Second, and philosophically more significant, *any* process which can be formalized so that it can be represented as series of instructions for the manipulation of discrete elements, can, at least in principle, be reproduced by such a machine. Thus even an analogue computer, provided that the relation of its input to its output can be described by a precise mathematical function, can be simulated on a digital machine.^{14*}

But such machines might have remained overgrown adding machines, had not Plato's vision, refined by two thousand years of metaphysics, found in them its fulfillment. At last here was a machine which operated according to syntactic rules, on bits of data. Moreover, the rules were built into the circuits of the machine. Once the machine was programmed there was no need for interpretation; no appeal to human intuition and judgment. This was just what Hobbes and Leibniz had ordered, and Martin Heidegger appropriately saw in cybernetics the culmination of the philosophical tradition.^{15*}

Thus while practical men like Eckert and Mauchly, at the University of Pennsylvania, were designing the first electronic digital machine, theorists, such as Turing, trying to understand the essence and capacity of such machines, became interested in an area which had thus far been the province of philosophers: the nature of reason itself.

In 1950, Turing wrote an influential article, "Computing Machinery and Intelligence," in which he points out that "the present interest in 'thinking machines' has been aroused by a particular kind of machine, usually called an 'electronic computer' or a 'digital computer.'" ¹⁶ He

then takes up the question “Can [such] machines think?”

To decide this question Turing proposes a test which he calls the imitation game:

The new form of the problem can be described in terms of a game which we call the “imitation game.” It is played with three people, a man (A), a woman (B), and an interrogator (C) who may be of either sex. The interrogator stays in a room apart from the other two. The object of the game for the interrogator is to determine which of the other two is the man and which is the woman. He knows them by labels X and Y, and at the end of the game he says either “X is A and Y is B” or “X is B and Y is A.” The interrogator is allowed to put questions to A and B thus:

C: Will X please tell me the length of his or her hair? Now suppose X is actually A, then A must answer. It is A’s object in the game to try to cause C to make the wrong identification. His answer might therefore be

“My hair is shingled, and the longest strands are about nine inches long.”

In order that tones of voice may not help the interrogator the answers should be written, or better still, typewritten. The ideal arrangement is to have a teleprinter communicating between the two rooms. Alternatively, the question and answers can be repeated by an intermediary. The object of the game for the third player (B) is to help the interrogator. The best strategy for her is probably to give truthful answers. She can add such things as “I am the woman, don’t listen to him!” to her answers, but it will avail nothing as the man can make similar remarks.

We now ask the question, “What will happen when a machine takes the part of A in this game?” Will the interrogator decide wrongly as often when the game is played like this as he does when the game is played between a man and a woman? These questions replace our original, “Can machines think?”¹⁷

This test has become known as the Turing Test. Philosophers may doubt whether merely behavioral similarity could ever give adequate ground for the attribution of intelligence,¹⁸ but as a goal for those actually trying to construct thinking machines, and as a criterion for critics to use in evaluating their work, Turing’s test was just what was needed.

Of course, no digital computer immediately volunteered or was drafted for Turing’s game. In spite of its speed, accuracy, and universality, the digital computer was still nothing more than a general-symbol manipulating device. The chips, however, were now down on the old Leibnizian bet. The time was ripe to produce the appropriate symbolism

and the detailed instructions by means of which the rules of reason could be incorporated in a computer program. Turing had grasped the possibility and provided the criterion for success, but his article ended with only the sketchiest suggestions about what to do next:

We may hope that machines will eventually compete with men in all purely intellectual fields. But which are the best ones to start with? Even this is a difficult decision. Many people think that a very abstract activity, like the playing of chess, would be best. It can also be maintained that it is best to provide the machine with the best sense organs that money can buy, and then teach it to understand and speak English. This process could follow the normal teaching of a child. Things would be pointed out and named, etc. Again I do not know what the right answer is, but I think both approaches should be tried.¹⁹

A technique was still needed for finding the rules which thinkers from Plato to Turing assumed must exist—a technique for converting any practical activity such as playing chess or learning a language into the set of instructions Leibniz called a theory. Immediately, as if following Turing's hints, work got under way on chess and language. The same year Turing wrote his article, Claude E. Shannon, the inventor of information theory, wrote an article on chess-playing machines in which he discussed the options facing someone trying to program a digital computer to play chess.

Investigating one particular line of play for 40 moves would be as bad as investigating all lines for just two moves. A suitable compromise would be to examine only the important possible variations—that is, forcing moves, captures and main threats—and carry out the investigation of the possible moves far enough to make the consequences of each fairly clear. It is possible to set up some rough criteria for selecting important variations, not as efficiently as a chess master, but sufficiently well to reduce the number of variations appreciably and thereby permit a deeper investigation of the moves actually considered.²⁰

Shannon did not write a chess program, but he believed that “an electronic computer programmed in this manner would play a fairly strong game at speeds comparable to human speeds.”²¹

In 1955 Allen Newell wrote a sober survey of the problems posed by the game of chess and suggestions as to how they might be met. Newell notes that “These [suggested] mechanisms are so complicated that it is

impossible to predict whether they will work.”²² The next year, however, brought startling success. A group at Los Alamos produced a program which played poor but legal chess on a reduced board. In a review of this work, Allen Newell, J. C. Shaw, and H. A. Simon concluded: “With very little in the way of complexity, we have at least entered the arena of human play—we can beat a beginner.”²³ And by 1957, Alex Bernstein had a program for the IBM 704 which played two “passable amateur games.”²⁴

Meanwhile, Anthony Oettinger was working on the other Turing line. Having already in 1952 programmed a machine which simulated simple conditioning, increasing or decreasing a set response on the basis of positive or negative reinforcement, Oettinger turned to the problem of language translation and programmed a Russian-English mechanical dictionary. Further research in these directions, it seemed, might lead to a computer which could be taught to associate words and objects.

But neither of these approaches offered anything like a general theory of intelligent behavior. What was needed were rules for converting any sort of intelligent activity into a set of instructions. At this point Herbert Simon and Allen Newell, analyzing the way a student proceeded to solve logic problems, noted that their subjects tended to use rules or shortcuts which were not universally correct, but which often helped, even if they sometimes failed. Such a rule of thumb might be, for example: always try to substitute a shorter expression for a longer one. Simon and Newell decided to try to simulate this practical intelligence. The term “heuristic program” was used to distinguish the resulting programs from programs which are guaranteed to work, so-called algorithmic programs which follow an exhaustive method to arrive at a solution, but which rapidly become unwieldy when dealing with practical problems.

This notion of a rule of practice provided a breakthrough for those looking for a way to program computers to exhibit general problem-solving behavior. Something of the excitement of this new idea vibrates in the first paragraph of Newell, Shaw, and Simon’s classic article “Empirical Explorations with the Logic Theory Machine: A Case Study in Heuristics.”

This is a case study in problem-solving, representing part of a program of research on complex information-processing systems. We have specified a system for finding proofs of theorems in elementary symbolic logic, and by programming a computer to these specifications, have obtained empirical data on the problem-solving process in elementary logic. The program is called the Logic Theory Machine (LT); it was devised to learn how it is possible to solve difficult problems such as proving mathematical theorems, discovering scientific laws from data, playing chess, or understanding the meaning of English prose.

The research reported here is aimed at understanding the complex processes (heuristics) that are effective in problem-solving. Hence, we are not interested in methods that guarantee solutions, but which require vast amounts of computation. Rather, we wish to understand how a mathematician, for example, is able to prove a theorem even though he does not know when he starts how, or if, he is going to succeed.²⁵

But Newell and Simon soon realized that even this approach was not general enough. The following year (1957) they sought to abstract the heuristics used in the logic machine, and apply them to a range of similar problems. This gave rise to a program called the *General Problem Solver* or *GPS*. The motivation and orientation of the work on the *General Problem Solver* are explained in Newell, Shaw, and Simon's first major report on the enterprise.

This paper . . . is part of an investigation into the extremely complex processes that are involved in intelligent, adaptive, and creative behavior. . . .

Many kinds of information can aid in solving problems: information may suggest the order in which possible solutions should be examined; it may rule out a whole class of solutions previously thought possible; it may provide a cheap test to distinguish likely from unlikely possibilities; and so on. All these kinds of information are *heuristics*—things that aid discovery. Heuristics seldom provide infallible guidance. . . . Often they “work,” but the results are variable and success is seldom guaranteed.²⁶

To convey a sense of the general heuristics their program employed, Newell and Simon introduced an example of everyday intelligent behavior:

I want to take my son to nursery school. What's the difference between what I have and what I want? One of distance. What changes distance? My automobile. My automobile won't work. What's needed to make it work? A new battery. What has new batteries? An auto repair shop. I want the repair shop to put in

a new battery; but the shop doesn't know I need one. What is the difficulty? One of communication. What allows communication? A telephone. . . . And so on.

This kind of analysis—classifying things in terms of the functions they serve, and oscillating among ends, functions required, and means that perform them—forms the basic system of heuristic of GPS. More precisely, this means-end system of heuristic assumes the following:

1. If an object is given that is not the desired one, differences will be detectable between the available object and the desired object.

2. Operators affect some features of their operands and leave others unchanged. Hence operators can be characterized by the changes they produce and can be used to try to eliminate differences between the objects to which they are applied and desired objects.

3. Some differences will prove more difficult to affect than others. It is profitable, therefore, to try to eliminate “difficult” differences, even at the cost of introducing new differences of lesser difficulty. This process can be repeated as long as progress is being made toward eliminating the more difficult differences.²⁷

With digital computers solving such problems as how to get three cannibals and three missionaries across a river without the cannibals eating the missionaries, it seemed that finally philosophical ambition had found the necessary technology: that the universal, high-speed computer had been given the rules for converting reasoning into reckoning. Simon and Newell sensed the importance of the moment and jubilantly announced that the era of intelligent machines was at hand.

We have begun to learn how to use computers to solve problems, where we do not have systematic and efficient computational algorithms. And we now know, at least in a limited area, not only how to program computers to perform such problem-solving activities successfully; we know also how to program computers to *learn* to do these things.

In short, we now have the elements of a theory of heuristic (as contrasted with algorithmic) problem solving; and we can use this theory both to understand human heuristic processes and to simulate such processes with digital computers. Intuition, insight, and learning are no longer exclusive possessions of humans: any large high-speed computer can be programmed to exhibit them also.²⁸

This field of research, dedicated to using digital computers to simulate intelligent behavior, soon came to be known as “artificial intelligence.” One should not be misled by the name. No doubt an artificial nervous system sufficiently like the human one, with other features such as sense

organs and a body, would be intelligent. But the term “artificial” does not mean that workers in artificial intelligence are trying to build an artificial man. Given the present state of physics, chemistry, and neurophysiology, such an undertaking is not feasible. Simon and the pioneers of artificial intelligence propose to produce something more limited: a heuristic program which will enable a digital information-processing machine to exhibit intelligence.

Likewise, the term “intelligence” can be misleading. No one expects the resulting robot to reproduce everything that counts as intelligent behavior in human beings. It need not, for example, be able to pick a good wife, or get across a busy street. It must only compete in the more objective and disembodied areas of human behavior, so as to be able to win at Turing’s game.

This limited objective of workers in artificial intelligence is just what gives such work its overwhelming significance. These last metaphysicians are staking everything on man’s ability to formalize his behavior; to bypass brain and body, and arrive, all the more surely, at the essence of rationality.

Computers have already brought about a technological revolution comparable to the Industrial Revolution. If Simon is right about the imminence of artificial intelligence, they are on the verge of creating an even greater conceptual revolution—a change in our understanding of man. Everyone senses the importance of this revolution, but we are so near the events that it is difficult to discern their significance. This much, however, is clear. Aristotle defined man as a rational animal, and since then reason has been held to be of the essence of man. If we are on the threshold of creating artificial intelligence we are about to see the triumph of a very special conception of reason. Indeed, if reason can be programmed into a computer, this will confirm an understanding of man as an object, which Western thinkers have been groping toward for two thousand years but which they only now have the tools to express and implement. The incarnation of this intuition will drastically change our understanding of ourselves. If, on the other hand, artificial intelligence should turn out to be impossible, then we will have to distinguish

human from artificial reason, and this too will radically change our view of ourselves. Thus the moment has come either to face the truth of the tradition's deepest intuition or to abandon the mechanical account of man's nature which has been gradually developing over the past two thousand years.

Although it is perhaps too early for a full answer, we must make an attempt to determine the scope and limits of the sort of reason which has come fully into force since the perfection of the "analytical engine." We must try to understand to what extent artificial intelligence is possible, and if there are limits to the possibility of computer simulation of intelligent behavior, we must determine those limits and their significance. What we learn about the limits of intelligence in computers will tell us something about the character and extent of human intelligence. What is required is nothing less than a critique of artificial reason.

II

The need for a critique of artificial reason is a special case of a general need for critical caution in the behavioral sciences. Chomsky remarks that in these sciences "there has been a natural but unfortunate tendency to 'extrapolate,' from the thimbleful of knowledge that has been attained in careful experimental work and rigorous data-processing, to issues of much wider significance and of great social concern." He concludes that

the experts have the responsibility of making clear the actual limits of their understanding and of the results they have so far achieved. A careful analysis of these limits will demonstrate that in virtually every domain of the social and behavioral sciences the results achieved to date will not support such "extrapolation."²⁹

Artificial intelligence, at first glance, seems to be a happy exception to this pessimistic principle. Every day we read that digital computers play chess, translate languages, recognize patterns, and will soon be able to take over our jobs. In fact this now seems like child's play. Literally! In a North American Newspaper Alliance release, dated December 1968, entitled "A Computer for Kids" we are told that

Cosmos, the West German publishing house . . . has come up with a new idea in gifts. . . . It's a genuine (if small) computer, and it costs around \$20. Battery operated, it looks like a portable typewriter. But it can be programmed like any big computer to translate foreign languages, diagnose illnesses, even provide a weather forecast.

And in a *Life* magazine article (Nov. 20, 1970) entitled "Meet Shakey, The First Electronic Person," the wide-eyed reader is told of a computer "made up of five major systems of circuitry that correspond quite closely to basic human faculties—sensation, reason, language, memory [and] ego." According to the article, this computer "sees," "understands," "learns," and, in general, has "demonstrated that machines can think." Several distinguished computer scientists are quoted as predicting that in from three to fifteen years "we will have a machine with the general intelligence of an average human being . . . and in a few months [thereafter] it will be at genius level. . . ."

The complete robot may be a few years off, of course, but anyone interested in the prospective situation at the turn of the century can see in the film *2001: A Space Odyssey* a robot named HAL who is cool, conversational, and very nearly omniscient and omnipotent. And this film is not simply science-fiction fantasy. *A Space Odyssey* was made with scrupulous documentation. The director, Stanley Kubrick, consulted the foremost computer specialists so as not to be misled as to what was at least remotely possible. Turing himself had in 1950 affirmed his belief that "at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted."³⁰ And the technical consultant for the film, Professor Marvin Minsky, working on an early prototype of HAL in his laboratory at M.I.T., assured Kubrick that Turing was, if anything, too pessimistic.

That Minsky was not misunderstood by Kubrick is clear from Minsky's editorial for *Science Journal*, which reads like the scenario for *2001*:

At first machines had simple claws. Soon they will have fantastically graceful articulations. Computers' eyes once could sense only a hole in a card. Now they recognize shapes on simple backgrounds. Soon they will rival man's analysis of

his environment. Computer programs once merely added columns of figures. Now they play games well, understand simple conversations, weigh many factors in decisions. What next?

Today, machines solve problems mainly according to the principles we build into them. Before long, we may learn how to set them to work upon the very special problem of improving their own capacity to solve problems. Once a certain threshold is passed, this could lead to a spiral of acceleration and it may be hard to perfect a reliable 'governor' to restrain it.³¹

It seems that there may be no limit to the range and brilliance of the properly programmed computer. It is no wonder that among philosophers of science one finds an assumption that machines can do everything people can do, followed by an attempt to interpret what this bodes for the philosophy of mind; while among moralists and theologians one finds a last-ditch retrenchment to such highly sophisticated behavior as moral choice, love, and creative discovery, claimed to be beyond the scope of any machine. Thinkers in both camps have failed to ask the preliminary question whether machines can in fact exhibit even elementary skills like playing games, solving simple problems, reading simple sentences and recognizing patterns, presumably because they are under the impression, fostered by the press and artificial-intelligence researchers such as Minsky, that the simple tasks and even some of the most difficult ones have already been or are about to be accomplished. To begin with, then, these claims must be examined.

It is fitting to begin with a prediction made by Herbert Simon in 1957 as his General Problem Solver seemed to be opening up the era of artificial intelligence:

It is not my aim to surprise or shock you. . . . But the simplest way I can summarize is to say that there are now in the world machines that think, that learn and that create. Moreover, their ability to do these things is going to increase rapidly until—in a visible future—the range of problems they can handle will be coextensive with the range to which the human mind has been applied.

Simon then predicts, among other things,

1. That within ten years a digital computer will be the world's chess champion, unless the rules bar it from competition.

2. That within ten years a digital computer will discover and prove an important new mathematical theorem.
3. That within ten years most theories in psychology will take the form of computer programs, or of qualitative statements about the characteristics of computer programs.³²

Unfortunately, the tenth anniversary of this historic talk went unnoticed, and workers in artificial intelligence did not, at any of their many national and international meetings, take time out from their progress reports to confront these predictions with the actual achievements. Now fourteen years have passed, and we are being warned that it may soon be difficult to control our robots. It is certainly high time to measure this original prophecy against reality.

Already in the five years following Simon's predictions, publications suggested that the first of Simon's forecasts had been half-realized, and that considerable progress had been made in fulfilling his second prediction. This latter, the theorem-discovery prediction, was "fulfilled" by W. R. Ashby (one of the leading authorities in the field) when, in a review of Feigenbaum and Feldman's anthology *Computers and Thought*, he hailed the mathematical power of the properly programmed computer: "Gelernter's theorem-proving program has discovered a new proof of the *pons asinorum* that demands no construction." This proof, Dr. Ashby goes on to say, is one which "the greatest mathematicians of 2000 years have failed to notice . . . which would have evoked the highest praise had it occurred."³³

The theorem sounds important, and the naive reader cannot help sharing Ashby's enthusiasm. A little research, however, reveals that the *pons asinorum*, or ass's bridge, is the elementary theorem proved in Euclidian geometry—namely that the opposite angles of an isosceles triangle are equal. Moreover, the first announcement of the "new" proof "discovered" by the machine is attributed to Pappus (A.D. 300).³⁴ There is a striking disparity between Ashby's excitement and the antiquity and simplicity of this proof. We are still a long way from "the important mathematical theorem" to be found by 1967.

The chess-playing story is more involved and might serve as a model for a study of the production of intellectual smog in this area. In 1958,

the year after Simon's prediction, Newell, Shaw, and Simon presented an elaborate chess-playing program. As described in their classic paper, "Chess-Playing Programs and the Problem of Complexity," their program was "not yet fully debugged," so that one "cannot say very much about the behavior of the program."³⁵ Still, it is clearly "good in [the] . . . opening."³⁶ This is the last detailed published report on the program. In the same year, however, Newell, Shaw, and Simon announced: "We have written a program that plays chess,"³⁷ and Simon, on the basis of this success, revised his earlier prediction:

In another place, we have predicted that within ten years a computer will discover and prove an important mathematical theorem. On the basis of our experience with the heuristics of logic and chess, we are willing to add the further prediction that only moderate extrapolation is required from the capacities of programs already in existence to achieve the additional problem-solving power needed for such simulation.³⁸

Public gullibility and Simon's enthusiasm was such that Newell, Shaw, and Simon's claims concerning their still bugged program were sufficient to launch the chess machine into the realm of scientific mythology. In 1959, Norbert Wiener, escalating the claim that the program was "good in the opening," informed the NYU Institute of Philosophy that "chess-playing machines as of now will counter the moves of a master game with the moves recognized as right in the text books, up to some point in the middle game."³⁹ In the same symposium, Michael Scriven moved from the ambiguous claim that "machines now play chess" to the positive assertion that "machines are already capable of a good game."⁴⁰

In fact, in its few recorded games, the Newell, Shaw, Simon program played poor but legal chess, and in its last official bout (October 1960) was beaten in 35 moves by a ten-year-old novice. Fact, however, had ceased to be relevant.

While their program was losing its five or six poor games—and the myth they had created was holding its own against masters in the middle game—Newell, Shaw, and Simon kept silent. When they speak again, three years later, they do not report their difficulties and disappointment. Rather, as if to take up where the myth left off, Simon published an article in *Behavioral Science* announcing a program which played

“highly creative” chess end games involving “combinations as difficult as any that have been recorded in chess history.”⁴¹ That the program restricts these end games to dependence on continuing checks, so that the number of relevant moves is greatly reduced, is mentioned but not emphasized. On the contrary, it is misleadingly implied that similar simple heuristics would account for master play even in the middle game.^{42*} Thus, the article gives the impression that the chess prediction is almost realized. With such progress, the chess championship may be *claimed* at any moment. Indeed, a Russian cyberneticist, upon hearing of Simon’s ten-year estimate, called it “conservative.”⁴³ And Fred Gruenberger at RAND suggested that a world champion is not enough—that we should aim for “a program which plays better than any man could.”⁴⁴ This regenerating confusion makes one think of the mythical French beast which is supposed to secrete the fog necessary for its own respiration.

Reality comes limping along behind these impressive pronouncements. Embarrassed by my exposé of the disparity between their enthusiasm and their results, AI workers finally produced a reasonably competent program. R. Greenblatt’s program called MacHack did in fact beat the author,^{45*} a rank amateur, and has been entered in several tournaments in which it won a few games. This limited success revived hopes and claims. Seymour Papert, the second in command at the M.I.T. robot project, leaped in to defend Simon’s prediction, asserting that “as a statement of what researchers in the field consider to be a possible goal for the near future, this is a reasonable statement.”⁴⁶ And on page 1 of the October 1968 issue of *Science Journal*, Donald Michie, the leader of England’s artificial intelligentsia, writes that “today machines can play chess at championship level.”⁴⁷ However, chess master de Groot, discussing the earlier chess programs, once said: “programs are still very poor chess players and I do not have much hope for substantial improvement in the future.” And another chess master, Eliot Hearst, discussing the M.I.T. program in *Psychology Today*, adds: “De Groot’s comment was made in 1964 and MacHack’s recent tournament showing would not require him to revise his opinion.”⁴⁸ Nor would most recent events. Greenblatt’s program has been gradually improved, but it seems to have

reached a point of saturation. During the past two years, it lost all games in the tournaments in which it had been entered, and received no further publicity. We shall soon see that given the limitations of digital computers this is just what one would expect.

It is to Greenblatt's credit that even in the heyday of MacHack he made no prediction; as for Simon and the world championship, the ten years are well up, and the computer is at best a class C amateur.^{49*}

This rapid rundown of the state of the art vis-à-vis two of Simon's predictions has, I hope, cleared the air. It is essential to be aware at the outset that despite predictions, press releases, films, and warnings, artificial intelligence is a promise and not an accomplished fact. Only then can we begin our examination of the actual state and future hopes of artificial intelligence at a sufficiently rudimentary level.

The field of artificial intelligence has many divisions and subdivisions, but the most important work can be classified into four areas: game playing, language translating, problem solving, and pattern recognition. We have already discussed the state of game-playing research. We shall now look at the work in the remaining three fields in detail. In Part I my general thesis will be that the field of artificial intelligence exhibits a recurring pattern: early, dramatic success followed by sudden unexpected difficulties. This pattern occurs in all four areas, in two phases each lasting roughly five years. The work from 1957 to 1962 (Chapter 1), is concerned primarily with Cognitive Simulation (CS)—the use of heuristic programs to simulate human behavior by attempting to reproduce the steps by which human beings actually proceed. The second period (Chapter 2) is predominantly devoted to semantic information processing. This is artificial intelligence in a narrower sense than I have been using the term thus far. AI (for this restricted sense I shall use the initials) is the attempt to simulate human intelligent behavior using programming techniques which need bear little or no resemblance to human mental processes. The difficulties confronting this approach have just begun to emerge. The task of the rest of Part I is to discover the underlying common source of all these seemingly unconnected setbacks.

These empirical difficulties, these failures to achieve predicted prog-

ress, never, however, discourage the researchers, whose optimism seems to grow with each failure. We therefore have to ask what assumptions underlie this persistent optimism in the face of repeated disappointments. Part II attempts to bring to light four deeply rooted assumptions or prejudices which mask the gravity of the current impasse, and to lay bare the conceptual confusion to which these prejudices give rise.

But these prejudices are so deeply rooted in our thinking that the only alternative to them seems to be an obscurantist rejection of the possibility of a science of human behavior. Part III attempts to answer this objection, insofar as it can be answered, by presenting an alternative to these traditional assumptions, drawing on the ideas of twentieth-century thinkers whose work is an implicit critique of artificial reason, although it has not before been read in this light.

We shall then, in the Conclusion, be in a position to characterize artificial reason and indicate its scope and limits. This in turn will enable us to distinguish among various forms of intelligent behavior and to judge to what extent each of these types of intelligent behavior is programmable in practice and in principle.

If the order of argument presented above and the tone of my opening remarks seem strangely polemical for an effort in philosophical analysis, I can only point out that, as we have already seen, artificial intelligence is a field in which the rhetorical presentation of results often substitutes for success, so that research papers resemble more a debater's brief than a scientific report. Such persuasive marshaling of facts can only be answered in kind. Thus the accusatory tone of Part I. In Part II, however, I have tried to remain as objective as possible in testing fundamental assumptions, although I know from experience that challenging these assumptions will produce reactions similar to those of an insecure believer when his faith is challenged.

For example, the year following the publication of my first investigation of work in artificial intelligence, the RAND Corporation held a meeting of experts in computer science to discuss, among other topics, my report. Only an "expurgated" transcript of this meeting has been released to the public, but even there the tone of paranoia which pervaded the discussion is present on almost every page. My report is called

“sinister,” “dishonest,” “hilariously funny,” and an “incredible misrepresentation of history.” When, at one point, Dr. J. C. R. Licklider, then of IBM, tried to come to the defense of my conclusion that work should be done on man-machine cooperation, Seymour Papert of M.I.T. responded:

I protest vehemently against crediting Dreyfus with any good. To state that you can associate yourself with one of his conclusions is unprincipled. Dreyfus’ concept of coupling men with machines is based on thorough misunderstanding of the problems and has nothing in common with any good statement that might go by the same words.⁵⁰

The causes of this panic-reaction should themselves be investigated, but that is a job for psychology, or the sociology of knowledge. However, in anticipation of the impending outrage I want to make absolutely clear from the outset that what I am criticizing is the implicit and explicit philosophical assumptions of Simon and Minsky and their co-workers, not their technical work. True, their philosophical prejudices and naïveté distort their own evaluation of their results, but this in no way detracts from the importance and value of their research on specific techniques such as list structures, and on more general problems such as data-base organization and access, compatibility theorems, and so forth. The fundamental ideas that they have contributed in these areas have not only made possible the limited achievements in artificial intelligence but have contributed to other more flourishing areas of computer science.

In some restricted ways even AI can have, and presumably will have practical value despite what I shall try to show are its fundamental limitations. (I restrict myself to AI because it is not clear that naïve Cognitive Simulation, as it is now practiced, can have any value at all, except perhaps as a striking demonstration of the fact that in behaving intelligently people do not process information like a heuristically programmed digital computer.) An artifact could replace men in some tasks—for example, those involved in exploring planets—without performing the way human beings would and without exhibiting human flexibility. Research in this area is not wasted or foolish, although a balanced view of what can and cannot be expected of such an artifact would certainly be aided by a little philosophical perspective.

